Vishnu Tripathi

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Summary

Published researcher with nearly 3 years of experience in AI/ML research, specializing in Computer Vision, Natural Language Processing, and Knowledge Systems. Author of 7 peer-reviewed publications across COLING 2025, ICIS 2024, PACIS 2025, and other international conferences. Demonstrated ability to bridge theoretical research and practical implementation, with expertise in deep learning, LLMs, and end-to-end system development. Experienced in leading cross-functional research teams and collaborating with academic and industry stakeholders.

Education

Master of Computer Application (Data Science) July 2023 – June 2025
Dev Sanskriti VishwaVidyalaya *Haridwar, Uttarakhand, India*

- Cumulative GPA: **8.85**

Bachelor Of Computer Application July 2020 – June 2023
Dev Sanskriti VishwaVidyalaya *Haridwar, Uttarakhand, India*

- Cumulative GPA: **8.22**

Experience

Junior Research Fellow (JRF) July 2025 – Present
CAIR Lab, Institute of Artificial Intelligence, DSVV *Haridwar, Uttarakhand*

- **Led development of an OCR system** for historic books with integrated image extraction capabilities, managing a **cross-functional team of 4 members** and leading weekly client meetings to align scope, timelines, and delivery.
- **Owned end-to-end project execution** for historical document digitisation, covering **requirements analysis, system architecture design, and stakeholder coordination**, delivering projects **on schedule and within agreed constraints.**
- **Published peer-reviewed research** with **3 papers accepted** at **AIHW 2025, PACIS 2025, and ACIS 2025**, contributing to applied research in **OCR, document analysis, and AI-driven digitisation.**

Research Intern December 2024 – June 2025
CAIR Lab, Institute of Artificial Intelligence, DSVV *Haridwar, Uttarakhand*
(Concurrent with development role at Sensekriti Innovation)

- Developed mass attendance system using deep learning-based facial recognition (ResNet50, GhostFaceNet), achieving 100% accuracy on 60-identity dataset with real-time processing capabilities in challenging lighting conditions.
- Designed knowledge graph extraction pipeline using Large Language Models for entity recognition and relationship extraction from unstructured documents, improving information retrieval efficiency and enabling semantic search.
- Proposed and validated novel methodologies for machine-generated text detection, achieving competitive performance in COLING 2025 competition on binary text classification.
- Published 4 peer-reviewed papers at ICIS 2024, COLING 2025, SIGGreen Pre-ICIS 2024, contributing frameworks for knowledge graphs, AI text detection, and sustainable reverse logistics.
- Conducted comprehensive literature reviews on OCR systems for historical documents, evaluating state-of-the-art approaches and identifying research gaps in degraded text recognition.

Full-Stack Developer & Applied Research August 2023 – June 2025
Sensekriti Innovation Pvt. Ltd. *Incubated at CAIR Lab, IAI, DSVV*
(Concurrent with research internship)

- Developed **production-grade** e-learning platform astrowala.world serving **300+ active users**, demonstrating practical application of research in real-world systems with measurable impact.
- Led client requirement gathering and stakeholder meetings, translating business needs into technical specifications and research-informed system design.
- Implemented cost-efficient cloud infrastructure architecture through innovative use of **Google Drive API** and **virtual machine deployment**.
- Designed security framework including session management, anomaly detection, and content protection mechanisms, ensuring reliable platform operations with 99%+ uptime.

Technical Skills

- **AI, Generative AI & Large Language Models (LLMs)**: Model Fine-tuning (LoRA, QLoRA), Multimodal Modelling, Prompt Engineering, Retrieval-Augmented Generation (RAG), Transformer Architectures, Deep Learning, LangChain, Hugging Face.
- **Machine Learning**: Supervised and Unsupervised Learning, Neural Networks, Model Training, Evaluation, and Optimisation, PyTorch, TensorFlow, Scikit-learn, spaCy, OpenCV, llama.cpp, ollama.
- **Research Engineering & MLOps**: CI/CD Pipelines (GitHub Actions, GitLab CI), Containerisation (Docker, Kubernetes), Automated Testing and Deployment.
- **Infrastructure & Cloud Computing**: Cloud Platforms: AWS, Azure (Virtual Machines).
- **Programming Languages**: Python, SQL, Javascript(ES6+), Java, C, C++
- **Frameworks, Tools & Systems**: LangChain, Git, Github, Ubuntu, ExcaliDraw, VSCode, Anaconda, Tailwind CSS, Express.js, Node.js, React.js, Flask.

Projects

Narrative Similarity: Hybrid Embedding + LLM Ensemble (SemEval 2026) | *Prompt Engineering, LL*

- Designed and evaluated **structured prompting strategies** for **Large Language Models (LLMs)** to solve **narrative similarity reasoning** tasks in SemEval 2026 Task 4.
- Implemented **few-shot and aspect-grounded prompting** that improved model reasoning performance from **70.5% to 77.5% accuracy** on the development dataset.
- Conducted **prompt sensitivity experiments**, discovering that **prompt framing and label priming** significantly influence model predictions and reasoning behaviour.
- Built a **hybrid ensemble system** combining **embedding similarity models** and **LLM-based reasoning models** using complementarity-driven model selection.
- Achieved **75% test accuracy** and ranked **3rd out of 47 international systems**, approaching the **human annotator ceiling of 78%**.

Knowledge Graph from Text Documents using LLM | *LLMs, Knowledge Graphs*

- Built a **knowledge graph system** from **unstructured text documents** using **Large Language Models (LLMs)** to extract **entities and relationships**, improving **information retrieval speed** and supporting more accurate decision-making.
- Implemented **entity recognition**, **relationship extraction**, and **graph visualisation**, delivering a clean, user-friendly interface that enabled intuitive exploration of complex document collections.
- Added **contextual navigation features** by linking **graph nodes and edges** directly to **source documents**, allowing users to jump to exact **PDF pages or sections**, reducing time spent searching for supporting evidence.

Mass Attendance System | *Computer Vision, Deep Learning*

- Built a **real-time facial recognition system** for **large-scale classroom attendance**, delivering **stable performance** in live classroom conditions with varying lighting, pose, and crowd density.
- Implemented **ResNet50** for **face detection** and **FaceNet512** for **feature extraction**, addressing **occlusions**, **tilted faces**, **lighting variation**, and **multiple orientations** to ensure reliable recognition.
- Validated the system in a **real-world deployment at ICFAI University (CSE Department)**, testing on **100 students** with a labelled dataset of **60 identities**, achieving **100% recognition accuracy** after **hardware and camera setup optimisation**.
- Designed a **data preprocessing and image filtering pipeline** to automatically select **high-quality facial samples** from noisy inputs, significantly improving **recognition consistency** across diverse data collection methods.

Astrowala.com Web App [***astrowala.world***] | *Web Development, Cloud Infrastructure*

- Developed and deployed a **production-ready learning platform** for astrology courses, supporting **automated course assignment**, **enrollment management**, and **content expiration**, actively used by **300+ students**.
- Designed a **cost-efficient infrastructure** using the **Google Drive API** mounted on a **Virtual Machine** for video delivery, achieving an **80% reduction in storage and streaming costs** compared to AWS S3 and Azure Blob.
- Implemented **platform security controls**, including **session-based authentication** with **single-device enforcement**, **geolocation-based anomaly detection**, and **automated email alerts**, strengthening content protection and user accountability.
- Owned **end-to-end production operations**, handling **maintenance**, **feature development**, **performance tuning**, and **database management**, with **MongoDB hosted locally** to ensure reliable and efficient data handling.

OCR on Historical Books | *OCR, Document Analysis*

- Conducted an **in-depth literature review** of **OCR systems for historical documents**, identifying **technical gaps**, **limitations**, and **best-fit approaches** for degraded and archival content.
- Evaluated and fine-tuned **multiple OCR pipelines**, adapting models for **low-quality scans**, **degraded text**, and **image artefacts**, resulting in **measurable gains in extraction accuracy**.
- Analysed and addressed **real-world edge cases**, including **noise**, **skewed pages**, **faded ink**, and **complex layouts**, improving robustness across diverse document types.
- Optimised **preprocessing techniques** and **threshold parameters** across the OCR pipeline to increase **text and image extraction precision** and reduce **manual correction effort**.

HistoAI – Ecosystem for Historical Books | *RAG, Knowledge Graphs, LLMs*

- Led **requirements gathering** and **recurring stakeholder reviews**, translating real-world needs into **technical specifications** and keeping development aligned with **changing business constraints**.
- Designed an **end-to-end system architecture** covering **structured data extraction**, **knowledge graph generation**, and a **RAG-based chatbot**, enabling **accurate, context-aware user queries**.
- Owned **coordination and delivery timelines**, managing **scope**, **priorities**, and **dependencies** to ensure **on-time delivery** despite evolving requirements.
- Supported **hands-on data execution** by analysing **real-world use cases**, conducting **targeted literature reviews**, running **experiments**, and delivering **plug-and-play solutions** to resolve **RAG chatbot performance challenges**.

Research Publications

- **Big Data-Guided Knowledge Graph and Its Visualization from Multi-Text Formats and Value-Added Interpretation** (2024).
  Research Paper - *International Conference on Information Systems 2024 Special Interest Group on Big Data Proceedings*.

- **AI-Monitors at GenAI Detection Task 1: Fast and Scalable Machine-Generated Text Detection** (2025).
  Workshop Paper - *Proceedings of the 1st Workshop on GenAI Content Detection (GenAIDetect), co-located with the International Conference on Computational Linguistics (COLING 2025)*.

- **AI-Driven Knowledge Graphs for Interoperable Digital Ecosystems Paper** (2025).
  Research Paper - *ACIS 2025 Proceedings.*

- **Information System Guided Reverse Logistics – Managing the Green Industry Environments** (2024).
  Research Paper -*SIGGreen Pre-ICIS 2024 Workshop, Special Interest Group on Green Information Systems.*

- **AI and IoHT guided Bioinformatics Solutions – Managing Stress Related Mental Health** (2025).
  Research Paper -*PACIS 2025 Proceedings, Track on IS for Healthcare.*

- **Sustainable Knowledge-Guided Big Data: Augmented AI Models for Predicting the Coexistence Between Diverse Digital Ecosystems** (2025).
  Research Paper - *International Conference on Information Systems (ICIS 2025), Special Interest Group on Big Data Proceedings*.

- **Explainable AI for Depression Detection from Social Media Texts using BERT and SHAP** (2025).
  Research Paper - *Presented at AIHW 2025, publication in progress.*

- **AI-Driven Virtual Health Assistants: Utilising Generative AI for Patient Engagement and Mental Health Support** (2025).
  Book Chpater - *Book Chapter, "Generative AI: Revolutionizing Creativity and Problem-Solving," Bentham Science Publishers, forthcoming.*

Certifications

- **ReactJS** – Udemy
- **Intermediate SQL** – DataCamp
- **Numpy for Data Science** – Udemy
- **Javascript** – HackerRank
- **Python** – HackerRank
- **HTML and CSS for Beginners From Basic to Advance** – Udemy
- **MERN stack BootCamp** – LearnCodeOnline

Scholastic Achievements

- **International Conference on Artificial Intelligence in Healthcare and Wellness** [*AIHW*]
  - Served as **First Author and Presenter**, representing **CAIR Lab, Institute of Artificial Intelligence, Dev Sanskriti Vishwavidyalaya, Haridwar**.
  - Proposed a **BERT-based depression detection framework** using **Reddit data** with **SHAP-based explainability**, emphasizing **model interpretability**, **clinical trust**, and **ethical responsibility** in mental health AI systems.

- **COLING2025 Competition on Binary Text Classification** [*COLING*]
  - Competed in **COLING's binary text classification challenge** to distinguish between **machine-generated** and **human-written text**, achieving **competitive performance**.

- Fine-tuned **Large Language Models** for **NLP classification tasks** and **co-authored a research paper** detailing methodology and experimental results.

- **2nd MBCC 2025 Conference at IIT Mandi** [*MBCC*]
  - Participated in the **International Conference on Mind, Brain, and Consciousness**, engaging in **interdisciplinary sessions** on Cognitive Psychology, Neuroscience, Indian Philosophy, Ayurveda, and **IKS-based mental health applications**.
  - Networked with **researchers from IITs, AIIMS, NIMHANS**, and **international universities** exploring consciousness studies and cognitive wellness.

- **International Conference on Scientific Spirituality and Consciousness (2024)** *ICSSC*
  - Volunteered at an **international conference** hosted by **Dev Sanskriti Vishwavidyalaya, Haridwar**, alongside my **master's studies**.
  - Assisted with **conference logistics**, **speaker coordination**, **registration management**, **session handling**, and supported **interdisciplinary discussions** on AI, consciousness studies, Indian Knowledge Systems (IKS), and scientific spirituality.

- **ICFAI Dehradun University Ideathon** [*Greenovation*]
  - Pitched an innovative green technology solution, effectively showcasing its potential impact and feasibility to judges and participants.

- **6th International Yoga Conference on AI/ML for Women's Health** [*Conference*]
  - Served on the **organizing committee**, coordinating sessions and workshops integrating **yoga and Ayurveda** with **ML/AI/DL applications** for **women's health**.

- **Tula's 24-Hour Hackathon** [*National Marathon Coding Competition*]
  - Developed a **full-stack e-office system** using the **MERN stack** to digitize paperwork in educational institutions, demonstrating **rapid prototyping** and **teamwork under time constraints**.

References

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